CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 99-039

WASTE DISCHARGE REQUIREMENTS
FOR
MR. EDGAR HUDSON
KINGS CANYON MOBILE HOME PARK
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Mr. Edgar Hudson (hereafter referred to as Discharger) owns and operates the Kings Canyon Mobile Home Park, Wastewater Treatment Facility (WWTF), which provides sewage service for the park residents.

2. The WWTF consists of a 0.035 mgd extended aeration package plant and two wastewater percolation/evaporation ponds connected in series (hereafter ponds).

3. Wastewater discharges from the WWTF are regulated by Waste Discharge Requirements Order No. 73-54 for Sims Mobile Home Park. Order No. 73-54 prescribes requirements for the reuse of 0.034 mgd of treated effluent on open areas within the Mobile Home Park. Order No. 73-54 is not adequate because it does not identify the current owner and WWTF operator, or accurately describe the method of disposal. Further, Order No. 73-54 is not consistent with the plans and policies of the Board.

4. The WWTF has been in operation since the park opened in 1973 and originally consisted of the package plant with a published design flow of 0.035 mgd and one 0.19 million gallon pond. During a 29 January 1979 inspection, Board staff observed the pond overflowing. To increase the WWTF's storage capacity, a second pond (hereafter east pond) with a capacity of 1.05 million gallons was constructed in early 1979. This pond was originally built without the use of engineered plans, and in August 1979 the Discharger hired consulting engineers to evaluate the structural integrity of the WWTF and it's adequacy in providing treatment and disposal for the Park's present and anticipated flows. The resulting report concludes that the package plant is probably capable of accepting and treating it's published design flow of 0.035 mgd, and estimates the two ponds have a percolative capacity of approximately 0.036 gpd. It does not account for precipitation. The report emphasizes the importance of regular maintenance of the ponds' bottoms in order to maintain their percolation characteristics and disposal capacities. The report provided a projected ultimate wastewater flow of 0.0317 mgd with all of the Park's 141 mobile home sites occupied.

5. In a 12 June 1992 letter to this office, the County of Fresno Department of Health reported effluent draining from the east pond through an overflow pipe to a surface water drainage course. The Discharger plugged the overflow pipe in 1995, but in subsequent

inspections Board staff observed the ponds near full during the summer and found physical evidence of overflows. On 13 December 1995, staff requested the Discharger to submit a technical report that included a water balance based on precipitation with a 100-year return frequency. In a 19 January 1996 meeting with the Discharger, Board staff agreed to suspend the request for an engineering report contingent upon the Discharger's ability to collect data on wastewater flows and pond level fluctuations. As part of this agreement, the Discharger placed freeboard markers in the ponds and installed a flow meter on the park's water supply. The Discharger has included pond freeboard measurements with monthly self monitoring reports. The measurements indicate that adequate freeboard has been maintained. During a 14 August 1997 inspection, Board staff observed the east pond with two feet of freeboard and the west pond dry, but found physical evidence that indicated the west pond had previously overflowed.

6. Based on the Discharger's Self Monitoring Reports for the period of 1996 to 1998, the WWTF effluent constituents have averaged as follows:

Constituent	Units	Effluent
BOD[1]	mg/l	9.9
Total Suspended Solids	mg/l	9.8
Settleable Solids	ml/l	0.05

[1] 5-day, 20°C BOD test.

7. The WWTF is in Section 32 of T13S, R26E, MDB&M, near the town of Dunlap, as shown in Attachment A, which is attached hereto and part of this Order by reference. The WWTF is at an elevation of approximately 1850 feet and is within a quarter mile of Little White Deer Creek. Surface flow from the site topography is to Little White Deer Creek, a tributary to Mill Creek, which flows into the Kings River approximately two miles below Pine Flat Dam. The WWTF lies within the Tulare Lake Hydrologic Basin, Kings River Hydrologic Unit, Dunlap Hydrologic Area, (No. 552.40), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

8. In 1995, the Board adopted a *Water Quality Control Plan, Second Edition, for the Tulare Lake Basin* (hereafter Basin Plan), which designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the Basin. These requirements implement the Basin Plan.

9. The beneficial uses of the Kings River from Pine Flat Dam to the Friant-Kern Canal are municipal and domestic supply, agricultural supply, hydropower generation, water contact recreation, non-contact water recreation, warm and cold freshwater habitat; spawning, reproduction and/or early development; wildlife habitat; freshwater replenishment; and groundwater recharge.

10. The beneficial uses of underlying groundwater are municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.

11. Soil depth to bedrock in the immediate vicinity of the ponds varies from a minimum of 7.0 feet to over 10 feet, and soils consist of a shallow topsoil overburden overlying decomposed granite.

12. The groundwater in the vicinity of the ponds is at least 15 feet below ground surface (bgs) based on information contained in the 8 August 1972, *Soil And Geological Report*.

13. The approximate annual average precipitation is 25 inches. Data from the Department of Water Resources Bulletin No. 195 indicates that the total annual precipitation having a 100-year return frequency is approximately 62 inches at nearby Balch Camp.

14. The WWTF site is bordered on the north by State Highway 180, on the west by Dunlap Road, on the south by a hillside covered with brush and oaks, and on the east by land covered with native grasses and occasional oaks.

15. The Discharger has indicated that, to date, sludge has not been removed from the WWTF. The pond bottoms have been disced but not excavated.

16. The permitted discharge is a previously permitted discharge that is consistent with the antidegradation provisions of the State Water Resources Control Board Resolution No. 68-16. The discharge is not expected to adversely affect groundwater.

17. The action to adopt waste discharge requirements for an existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, California Code of Regulations (CCR), Section 15301.

18. The Board has notified the Discharger and interested agencies and persons of it's intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

19. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 73-54 is rescinded and that Mr. Edgar Hudson and his agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at Kings Canyon Mobile Home Park:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially treated waste is prohibited.

3. Discharge of waste classified as 'hazardous,' as defined in Section 2521(a) of Title 23, CCR, Section 2510, et seq., (hereinafter Chapter 15), or ‘designated,’ as defined in Section 13173 of the California Water Code, is prohibited.

4. Discharge of wastes to areas other than the ponds described in Finding Nos. 2 and 7 is prohibited.

B. Discharge Specifications:

1. The monthly average daily discharge flow from the package plant shall not exceed 0.034 million gallons.

2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

3. As a means of discerning compliance with Discharge Specification No. B.2 , the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.

4. The effluent from the WWTF shall not exceed the following limits:

Constituent	Units	Monthly Average	Maximum
BOD_5[1]	mg/l	40	80
Total Suspended Solids	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5

[1] 5-day, 20°C BOD test.

5. Percolation ponds shall not have a pH less than 6.5 or greater than 9.0.

6. Percolation ponds shall be managed to prevent breeding of mosquitoes. In particular:

   a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

   b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.

   c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

7. Public contact with wastewater in the treatment and disposal areas shall be precluded through such means as fences and signs, or acceptable alternatives.

8. The WWTF shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

9. Percolation ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the winter. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet in any pond (measured vertically).

10. On or about 1 October of each year, one of the two ponds shall be empty and the bottom sufficiently dry to allow for pond bottom maintenance. The pond emptied shall alternate every year such that each pond will receive service every other year. Maintenance shall consist of discing pond bottoms and, as necessary, excavating to restore design depth and percolation rate.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, California Code of Regulations, Division 2, Subdivision 1, Section 20005, et seq.

2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 60 days in advance of the change.

3. Use and disposal of sewage sludge shall comply with existing federal, state, and local laws, regulations, and ordinances, including permitting requirements and technical standards included in 40 CFR 503.

   If the State Water Resources Control Board and the Regional Water Quality Control Boards assume primacy to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background groundwater quality.

E. Provisions:

1. The Discharger shall comply with Monitoring and Reporting Program No. 99-039, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

3. Disposal capacity at conditions specified in this Order is unproven. At greater flow, the Discharger may be required to submit a water balance which demonstrates that evaporation/percolation ponds have sufficient capacity to comply with Discharge Specification B.9. The report shall be prepared by a California registered engineer with experience in wastewater disposal. Failure to demonstrate necessary disposal capacity will be cause to open and revise the Order to reflect a lower allowable capacity.

4. The Discharger shall notify the Board **at least 120 days prior to** any planned physical alterations or additions to the WWTF.

5. The Discharger shall use the best practicable control technique currently available to comply with this Order.

6. A copy of this Order shall be kept at the Mobile Home Park and shall be available for reference by the WWTF operator, who shall be familiar with it's contents.

7. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

   To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the proposed owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

8. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

9. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 30 April 1999.

GARY M. CARLTON, Executive Officer

CTS:fmc:4/30/99

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 99-039
FOR
MR. EDGAR HUDSON
KINGS CANYON MOBILE HOME PARK
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff. A description of the stations shall be submitted to Board and attached to this Program.

INFLUENT MONITORING

Influent monitoring shall include at least the following:

Constituent	Units	Type of Sample	Sampling Frequency
Daily Flow	Gals/day	Estimate[1]	Daily

[1] WWTF flow shall be estimated from readings taken from the park's water supply flow meter.

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the percolation ponds. Effluent samples shall be representative of the volume and nature of the discharge. Time of collection of grab samples shall be recorded. Effluent monitoring shall include at least the following:

Constituents	Units	Type of Sample	Sampling Frequency
BOD_5[1]	mg/l	Grab	2/Month[3]
Total Suspended Solids	mg/l	Grab	2/Month[3]
Settleable Solids	ml/l	Grab	2/Month[3]
EC[2]	µmhos/cm	Grab	Annual

[1] Five day, 20☜C biological oxygen demand test
[2] Conductivity @ 25^o C.
[3] If results indicate violation of the effluent limitation for the constituent, the frequency of monitoring shall be increased to weekly for the duration of the period of noncompliance.

POND MONITORING

Percolation ponds shall be monitored as follows:

Constituent	Unit	Type of Sample	Frequency
pH	pH Units	Grab	Monthly
Freeboard	feet	Observation	Weekly
Dissolved Oxygen[1]	mg/l	Grab[1]	Weekly[2]

1 Samples shall be collected at a depth of one foot, opposite the inlet between 0800 and 0900 hours.
2 If results indicate a concentration of less than 1.0 mg/l or if offensive odors are noted, the frequency of monitoring shall be increased as necessary to characterize the period of non-compliance.

Permanent markers shall be present in percolation ponds with calibrations indicating the water level at design capacity and available operational freeboard. In addition, the Discharger shall inspect the condition of the ponds once per week and include visual observations in the monthly self monitoring report. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether burrowing animals or insects are present; and the color of the pond (e.g. dark sparkling green, dull green, yellow, gray, tan, brown, etc.).

REPORTING

Daily, weekly, 2/month, and monthly monitoring data shall be included in monthly monitoring reports submitted to the Board by the **20th day** of the month following sample collection. Completion of the yearly pond maintenance required in Discharge Specification B.10 shall be noted on the monitoring report for the month in which it was performed.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

By **20 January of each year,** the Discharger shall submit a written Annual Report to the Executive Officer containing the following:

a. The names, titles, certificate grade, if any, and general responsibilities of persons operating and maintaining the WWTF.

b. The names and telephone numbers of persons to contact regarding the WWTF for emergency and routine situations.

c. The total quantity of sludge disposed of during the previous year, if any, and the ultimate disposal site(s).

d. A pond maintenance report describing which pond was serviced, when it was serviced, and the type of maintenance performed.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

Ordered by:
GARY M. CARLTON, Executive Officer

30 April 1999
(Date)

CTS:fmc:4/30/99

INFORMATION SHEET

MR. EDGAR HUDSON
ORDER NO. 99-039
KINGS CANYON MOBILE HOME PARK
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

The WWTF is an extended aeration package plant with a published capacity of 35,000 gallons per day for wastewater having an average BOD_5 of 240 mg/l.

The Kings Canyon Mobile Home Park contains spaces for 141 mobile homes. Using an assumed flow of 225 gallons per day (gpd) for each mobile home, the ultimate wastewater flow for full park occupancy has been estimated to be 31,725 gpd.

The disposal system originally consisted of a single 0.19 million gallon evaporation/percolation pond (west pond) but was expanded with the addition of a second 1.05 million gallon evaporation/percolation pond (east pond) in 1979. The ponds were originally connected in series but are supposed to now contain piping to allow the option of diverting effluent to one pond or the other.

An engineering report prepared after construction of the east pond in 1979, indicates soils in the immediate vicinity of the east and west ponds are comprised of fine to medium grain decomposed granite overlying bedrock at a depth of 7-feet to over 10-feet. A soil and geologic report prepared in 1972 concludes that the bedrock is likely fractured and that groundwater in this same area is at least 15 to 25 feet below ground surface. In the vicinity of intermittent Little White Deer Creek, several hundred yards southeast of the ponds, groundwater is very close to the surface.

An engineering report that evaluated WWTF treatment and storage capacity was submitted to this office shortly after construction of the east pond in 1979. Based on flow estimates for the period of 1973 to 1978 and the size of the original pond, a percolation rate estimate of 1.94 inches per day was concluded in the report. This equates to a total disposal capacity of 36,000 gpd for the two ponds. The report emphasized, however, that this disposal rate could only be maintained by periodically removing a pond from service, drying it, and discing or scraping the bottom. While the Park's full occupancy wastewater flow is estimated to be less, the Discharger has requested that the flow limit remain at 0.034 mgd, as in previous Order No. 73-54. The conclusions of the 1979 report support this flow limit.

Because the WWTF does not provide disinfection, reclamation of wastewater through landscape irrigation has been precluded from this order.

The WWTF is a small facility with evaporation/percolation disposal but is not considered remote due to it's proximity to the park's residents. Effluent limits in this Order reflect current Board policy for regulating a WWTF of this type and implement the *Water Quality Control Plan for the Tulare Lake Basin, Second Edition - 1995*.

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